9 Work and Energy

We use force while pulling or pushing to move objects around us. This is called doing work. We need energy to be able to do any work. We will learn about the concepts of work and energy in this chapter.

Learn about

- Work
- Energy
- Types of energy
- Renewable and non-renewable energy

Work

***Work** is said to be done when a force applied on an object causes it to move a certain distance in the direction of the force.* There are also many situations in our daily life when we feel tired after performing a certain task, but no work may have actually been done.

Examples of Work Done

The following are some examples of work being done:

- Pulling or pushing a cart through some distance
- Carrying a load up a staircase
- Pushing and moving a box up a ramp
- Running or jogging
- Playing outdoor sports such as football or badminton
- Swimming

Pushing a cart

Examples of Work Not Being Done

The following are some examples when work is not being done:

- Reading a book or studying for an examination
- Holding a heavy load, but not moving
- Pushing a heavy box or wall that does not move

Reading a book

Holding a heavy load

Although reading a book involves a lot of mental work, it will not be considered as physical work. Similarly, if someone holds a heavy load but does not move through a distance, no work is considered to be done by that person. If you try to push a wall, you might get tired after some time, but you have not done any work because the wall does not move. Therefore, work is said to be done only when the object on which the force is applied moves a certain distance in the direction of the force.

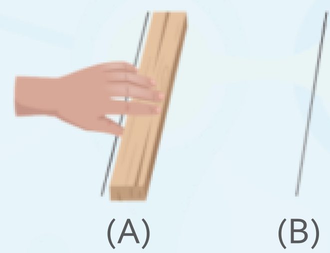
Activity

Aim: To show work being done (Situation 1)

Materials required: A table, a small block of wood, a ruler, and a piece of chalk

Observation:

1. Using the chalk, draw two lines some distance apart on the table.
2. Mark them as A and B.
3. Place the block of wood on position A.
4. Apply force and push the block of wood with your hand from position A to position B.



Observation: By pushing the block of wood, we have moved it some distance away from its original position.

Conclusion: The force of our hand has displaced the block of wood by some distance. Hence, work is done.

Activity

Aim: To show work being done (Situation 2)

Materials required: A piece of paper, a tub, a bucket of water, a mug, and a table

Procedure:

1. Place the tub on the table.
2. Pour water into the tub with the help of the mug from the bucket.
3. Make a paper boat and gently place it on the surface of the water in the tub.
4. Switch off the ceiling fan in the room.
5. Now, blow air at one end of the boat.

Observation: The boat will start moving from its original position.

Conclusion: The force of blowing air has moved the boat from its original position. Hence, work is done.

Energy

Work and energy are closely related to each other. ***Energy** is defined as the capacity or ability to do work.* Whenever work is done, there is always a change or transfer of energy. Thus, energy is needed to do work.

We get energy to do work from food items that are rich in carbohydrates and fats. These are also called **energy-giving foods**. Cereals (e.g., rice, wheat, and corn), fruits, sugar, potato, and small quantities of oil, butter, cheese, *ghee*, cream, and dry fruits are some examples of energy-giving foods. The amount of energy that we need also depends on the task that we are doing. For example:

- We need more energy to swim or jog than what we need to walk slowly.
- We need more energy to climb a staircase than what we need to walk a few steps on a plain ground.
- We need more energy to play sports such as football, cricket, and badminton than what we need to sit at home and watch these sports on the television.

The water stored behind a dam has energy.

Types of Energy

There are different forms of energy. Let us learn about them.

Mechanical Energy

*The energy that an object possesses due to its position or its movement is called **mechanical energy**.* The water stored at a height behind a dam in a reservoir[1] has energy. It can be used to rotate turbines and produce electricity when water falls down from above.

Burning of fuels releases heat energy.

Heat Energy

Burning of fuels such as LPG (liquefied petroleum gas) releases energy in the form of heat that can be used for cooking food. Heat produced by burning of coal can rotate turbines and produce electricity. The sun is the natural source of heat energy on Earth. The heat energy of the sun helps us to dry our clothes.

Fact File

Cattle dung is used for producing biogas, which is used for domestic cooking in villages.

[1]**reservoir:** a natural or artificial lake where water is stored

Electrical Energy

Electrical energy is generated at power stations and supplied to our homes. This form of energy is required to run appliances such as radios, televisions, and computers.

Appliances such as televisions and computers use electrical energy.

Wind Energy

The energy present in moving air (wind) is called ***wind energy****.* This energy is converted into electrical energy by windmills.

Electric bulbs, tube lights, and candles are some sources of light energy.

Light Energy

Light energy helps us to see things around us. The sun is the natural source of light energy on Earth. Other than the sun, tube lights, bulbs, and candles are also sources of light energy.

Solar Energy

The energy obtained from the rays of the sun is called ***solar energy****.* We also get light and heat energy from the sun.

Solar energy

Sound Energy

Sound energy is produced when an object vibrates[2]. This sound energy can travel through air, water, wood, or metal and reach our ears. For example, when the bell present inside an alarm clock vibrates, it produces sound energy. This sound energy travels through the air and reaches our ears. Because of this, we can hear the alarm of the clock ringing.

Vibrating strings generate sound energy.

[2]**vibrates:** to move to and fro continuously

Activity

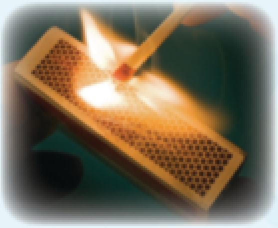
SCIENTIFIC SKILLS EXPERIENTIAL LEARNING

Aim: To demonstrate heat energy

Materials required: A small piece of paper, a pair of tongs, a candle, a matchbox, and a small table

Procedure: [**Note: Adult supervision required**]

1. Place the candle on the table.
2. Light it with a match.
3. Hold the piece of paper over the flame of the candle with the help of the tongs.



Heat energy

Observation: The paper catches fire easily and is burnt quickly.

Conclusion: Heat is a form of energy. Candle flame contains heat energy because of which the paper gets burnt.

Activity

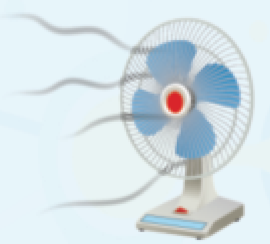
SCIENTIFIC SKILLS EXPERIENTIAL LEARNING

Aim: To demonstrate wind as a form of energy

Materials required: A small electric table fan, an electric socket, a small piece of cotton, and a table

Procedure: [**Note: Adult supervision required**]

1. Place the table fan on the table.
2. Fix the plug to the electric socket.
3. Place the piece of cotton some distance away from the fan, on the table.
4. Now, switch on the fan.



Wind energy

Observation: The blades of the fan start moving very fast, and simultaneously, the cotton is blown away from the table.

Conclusion: The wind produced by the moving blades of the fan blows away the piece of cotton. Thus, wind is a form of energy.

Difference between Work and Energy

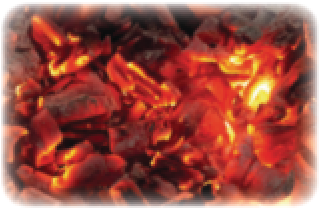
Work is said to be done when a force applied on an object causes that object to move through a distance in the direction of the force. But to be able to do work, we need to spend some energy. **Energy** is defined as the capacity or ability to do work. For example, to throw a stone, we need energy. Using this energy, we can throw the stone. The act of throwing the stone is the work done by us.

Renewable and Non-renewable Energy

- *The energy sources that last for a long period of time, such as sun, water, and wind, are called* **renewable energy sources**. The energy we get from these sources is known as **renewable energy**. Solar energy, hydro energy (energy of moving water), and wind energy are examples of renewable energy.
- *The energy sources present in a limited amount in nature which do not last for a long period of time are called* **non-renewable energy sources**. Examples of such sources are coal, petroleum, and natural gas. The energy that we get from these sources is known as **non-renewable energy**. They cannot be easily made by natural means, so we should use them judiciously[3] and carefully.

Windmills use wind energy to generate electricity.

Solar panels use solar energy to generate electricity.



Coal is a source of non-renewable energy.

Petroleum is a source of non-renewable energy.

Renewable and non-renewable energy

Know Your SDGs

SDG 7: **Affordable and Clean Energy**

(*Ensure access to affordable, reliable, sustainable and modern energy for all*)

Everyone deserves access to clean energy. What are some challenges that countries might face in bringing clean energy sources to everyone?

Questions

UNDERSTANDING

Mention whether work is being done or not in the following activities.

1. When you push a cart in the grocery store, and it moves ....
2. When you walk in a park ....
3. When you stand still and wait for the school bus at the bus stand ....
4. When you push a heavy box, and it does not move ....

[3]judiciously: wisely

Wrap Up

- Work is said to be done when a force applied on an object causes that object to move through a distance, in the direction of the force.
- Energy is defined as the capacity or ability to do work.
- Whenever work is done, there is always a change or transfer of energy.
- There are different forms of energy such as mechanical energy, heat energy, electrical energy, wind energy, light energy, solar energy, and sound energy.
- The energy sources that last for a long period of time, such as sun, water, and wind, are called renewable energy sources. The energy we get from these sources is known as renewable energy.
- The energy sources that do not last for a long period of time, such as coal, petroleum, and natural gas, are called non-renewable energy sources. The energy we get from these sources is known as non-renewable energy.

Exercises

REMEMBERING UNDERSTANDING APPLYING

SECTION I

A. Choose the correct option.

1. Work is done while ....
   a. swimming b. reading a book
   c. sleeping d. holding a heavy load but not moving
2. Which of the following actions does not involve work being done, even though effort might be exerted?
   a. Running b. Studying c. Walking d. Playing cricket
3. Which of the following options includes foods that are known to provide energy to the body?
   a. Wheat b. Oil c. Corn d. All of these
4. Which of the following sources can provide heat energy?
   a. LPG b. Coal c. Sun d. All of these
5. Your school is planning to install a new, environmentally friendly energy system. Which of the following energy sources should they choose to ensure it can be renewed naturally?
   a. Petroleum b. Sun c. Natural gas d. Coal

B Assertion and Reasoning questions

1. **Assertion (A):** A person pushing a wall with no movement is not doing any work.

   Reason (R): The force applied by the person is not enough to overcome the resistance of the wall.

   a. Both A and R are True b. Both A and R are False

   c. A is True and R is False d. A is False and R is True

2. **Assertion (A):** Wind energy cannot be converted into electrical energy.

   Reason (R): Wind energy is the energy present in moving air.

   a. Both A and R are True b. Both A and R are False

   c. A is True and R is False d. A is False and R is True

C Choose the correct option to fill in the blank.

1. When we are .... (jogging/sleeping), work is done.
2. Energy is the .... (capacity/thinking) to do work.
3. Whenever work is done, there is .... (no transfer/transfer) of energy.
4. The sun is a/an .... (artificial/natural) source of heat energy.
5. Wind energy is converted into electrical energy by .... (windmills/electric bulbs).

D Give two examples of the following.

1. Fuels that release heat energy ....
2. Appliances that run on electrical energy ....
3. Objects that produces sound energy ....
4. Renewable energy sources ....
5. Non-renewable energy sources ....

SECTION II

REMEMBERING UNDERSTANDING APPLYING

E Short answer questions

1. Define work with an example.
2. Give an example of a situation where force is applied without doing work.
3. Under what circumstances is work said to be done?

4. After playing a long game of football, you feel tired and hungry. List five energy-giving foods you could eat to restore your energy and explain why these foods are effective.
5. Write the names of any five types of energy.

F Long answer questions

1. Compare and contrast two scenarios: (a) You are pushing a car that moves forward, and (b) you are pushing against a solid wall that does not move. Discuss why work is done in one scenario and not in the other.
2. What is the difference between work and energy? Explain with an example.
3. How can we say that the amount of energy we need depends on the task that we are doing?
4. Differentiate between renewable and non-renewable energy.

REMEMBERING UNDERSTANDING APPLYING ANALYSING

Picture Study

Observe the given picture and answer the following questions.

1. Name the given structure shown in the picture.

....

2. Which energy does it use?

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....

3. Is it renewable or non-renewable energy? How can you say so?

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....

4. Is this structure beneficial for us? Write its role.

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....

....

My Learning Corner

TP ANALYSING EVALUATING CREATING

A ### Think about

SCIENTIFIC SKILLS

1. Suppose you are holding your school bag on your shoulder, standing still, and waiting for your school bus to arrive. Are you doing work? [Hint: You are 'standing still'. Think and answer.]

....

....

B ### Try out

VALUES

1. What are the different forms of energy that you occasionally use at your home? Make a list of them in your Science scrapbook. Also, mention the different ways in which you can save energy at home.

KNOWLEDGE OF INDIA

SUBJECT INTEGRATION

2. Collect information from magazines, newspapers, and the Internet regarding how electricity is produced in India in various power plants (hydro, wind, nuclear, and thermal). On the political map of India, mark any ten important power plants where electricity is generated. Use H, W, N, and T for the types of power plant—hydro, wind, nuclear, and thermal, respectively.

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• discuss the meaning of work by citing examples from daily life.	
• explain situations where work is being done/not being done.	
• demonstrate through activities how work is done/not done in different situations.	
• explain why energy is needed for work.	
• explain the importance of energy in daily life.	
• differentiate between work and energy with examples.	
• list the different forms of energy (light, electrical, heat, and sound) with examples for each.	
• differentiate between renewable and non-renewable energy with examples.	

Worksheet

Search and encircle the names of seven types of energy, in the Energy Grid given below. Also, write the definition and a few lines related to them in the space provided.

M	O	A	P	R	G	C	Q	H	S
Y	H	O	E	M	A	W	V	E	O
M	E	C	H	A	N	I	C	A	L
X	L	I	G	H	T	N	W	T	A
Z	E	S	O	U	N	D	Q	I	R
T	C	Y	N	Q	B	H	Z	O	F
K	T	N	A	Z	U	C	S	G	T
E	R	V	T	Y	B	N	Z	R	I
R	I	L	Q	N	R	I	F	P	Z
O	C	W	U	N	Q	C	P	A	M
Q	A	N	R	P	Z	U	J	E	B
Z	L	B	U	I	Q	Z	V	W	Q

Type of energy: Definition and details

1. ....

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2. ....

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3. ....

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4. ....

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5. ....

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6. ....

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7. ....

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